



Lakshya JEE 2.0 (2024)

Method of Differentiation

DPP-03

1. The derivative of e^{x^3} with respect to $\log x$ is

- (1) e^{x^3} (2) $3x^2 2e^{x^3}$
 (3) $3x^3 e^{x^3}$ (4) $3x^2 e^{x^3} + 3x^2$

2. Let $y = t^{10} + 1$ and $x = t^8 + 1$, then $\frac{d^2 y}{dx^2}$ is:

- (1) $\frac{5}{2}t$ (2) $20t^8$
 (3) $\frac{5}{16t^6}$ (4) None of these

3. If $x = \log p$ and $y = \frac{1}{p}$, then:

- (1) $\frac{d^2 y}{dx^2} - 2p = 0$ (2) $\frac{d^2 y}{dx^2} + y = 0$
 (3) $\frac{d^2 y}{dx^2} + \frac{dy}{dx} = 0$ (4) $\frac{d^2 y}{dx^2} - \frac{dy}{dx} = 0$

4. Differentiate $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t. $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

- (1) 0 (2) 1
 (3) -1 (4) $\tan x$

5. If $x = \tan^{-1}\left(\frac{\sqrt{1+\sin t} + \sqrt{1-\sin t}}{\sqrt{1+\sin t} - \sqrt{1-\sin t}}\right)$

and $y = \tan^{-1}\left(\frac{\sqrt{1+t^2}-1}{t}\right)$, then $\frac{dy}{dx} =$

- (1) $\frac{1}{t^2 + 1}$ (2) $t^2 + 1$
 (3) $\frac{1}{t^2 - 1}$ (4) $t^2 - 1$

6. If $y = A \cos(\log x) + B \sin(\log x)$, then

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y =$$

- (1) 1 (2) 0
 (3) -1 (4) 2

7. Differentiate $f(\sin x)$ w.r.t. $f(\cos x)$

- (1) $-\frac{f'(\sin x) \cos x}{f'(\cos x) \sin x}$
 (2) $\frac{f'(\cos x) \sin x}{f'(\sin x) \cos x}$
 (3) $-\frac{f'(\cos x) \sin x}{f'(\sin x) \cos x}$
 (4) $\frac{f'(\cos x) \sin x}{f'(-\sin x) \cos x}$

8. If, $y = \sin x + e^x$, then $\frac{d^2 x}{dy^2} =$

- (1) $(-\sin x + e^x)^{-1}$ (2) $\frac{\sin x - e^x}{(\cos x + e^x)^2}$
 (3) $\frac{\sin x - e^x}{(\cos x + e^x)^3}$ (4) $\frac{\sin x + e^x}{(\cos x + e^x)^3}$

9. $x = t \cos t$, $y = t + \sin t$, then $\frac{d^2 x}{dy^2}$ at $t = \frac{\pi}{2}$ is:

- (1) $\frac{\pi+4}{2}$ (2) $-\frac{\pi+4}{2}$
 (3) -2 (4) None of these

10. If $y = \cos^2 \frac{3x}{2} - \sin^2 \frac{3x}{2}$, then $\frac{d^2 y}{dx^2}$ is

- (1) $-3\sqrt{1-y^2}$ (2) $9y$
 (3) $-9y$ (4) $3\sqrt{1-y^2}$

11. If $f(x) = \cos^{-1}\left\{\frac{1-(\log_e x)^2}{1+(\log_e x)^2}\right\}$, then $f'(e)$

- (1) Does not exist
 (2) Is equal to $\frac{2}{e}$
 (3) Is equal to $\frac{1}{e}$
 (4) Is equal to 1

12. If $x = 2\sin\theta - \sin 2\theta$ and $y = 2\cos\theta - \cos 2\theta$,

$\theta \in [0, 2\pi]$, then $\frac{d^2y}{dx^2}$ at $\theta = \pi$ is:

- (1) $\frac{3}{8}$ (2) $\frac{3}{2}$
(3) $\frac{3}{4}$ (4) $-\frac{3}{4}$

13. The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect

to $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at $x = \frac{1}{2}$ is:

- (1) $\frac{2\sqrt{3}}{3}$ (2) $\frac{2\sqrt{3}}{5}$
(3) $\frac{\sqrt{3}}{10}$ (4) $\frac{\sqrt{3}}{12}$

14. Let $x^k + y^k = a^k$ where $a, k > 0$ and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$
then find k

- (1) $\frac{1}{3}$ (2) $\frac{2}{3}$
(3) $\frac{4}{3}$ (4) 2

15. If $y = (1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n})$, then the value of $\frac{dy}{dx}$ at $x = 0$ is

- (1) 0 (2) -1
(3) 1 (4) 2

16. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, then $\frac{dy}{dx}$ equals

- (1) $\sqrt{(1-x^2)(1-y^2)}$ (2) $\sqrt{\frac{1-y^2}{1-x^2}}$
(3) $\sqrt{\frac{1-x^2}{1-y^2}}$ (4) None of these



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (3)
3. (3)
4. (2)
5. (3)
6. (2)
7. (1)
8. (3)

9. (2)
10. (3)
11. (3)
12. (1)
13. (3)
14. (2)
15. (3)
16. (2)



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